

Offline, once: metric mapping + instance registration

Multiview capture → COLMAP + metric alignment → 3D Gaussian Splatting → SAM + cross-view association → Persistent IDs + one-time naming

Wearer gaze → world fixations



Localize camera

ArUco map frame



Lift gaze to 3D

Pose + surface depth



Form fixation events

Cluster in space + time



World-space fixation events

3DGS-based instance disambiguation

Visible surfaces within gaze uncertainty

Map view at wearer pose



Gaze + angular uncertainty

Local query



Depth + opacity + ID

Visible mass

$$q_k = m_k / W$$

Size correction

$$q_k / C_k$$

Rank + gate

ID / no binding

Visibility · uncertainty · target angular size

Shared 3DGS instance map

One metric frame for user and robot



Pose-conditioned depth + opacity

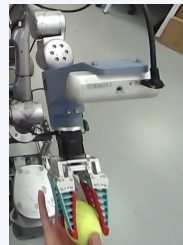
Persistent IDs + metric locations

Robot execution

Grounded target → action

Map-frame navigation

Target re-observation



Grasp & deliver

Speech → grounded commands

Streaming ASR

Word timestamps

LLM: actions + slots

"Bring me that ball."

Word-fixation binding

gaze
word

ball_R

"that"

Compile + dispatch

Action + ID + world parameters

On successful binding

Execute commands

move · pick · place